

Department of Mechanical Engineering
B.Tech Project Presentation on

VOLLEY: Controlled CubeSat Deployment from a Hosted Orbital Platform



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Final review: 10 October 2026

SYMBIOSIS INSTITUTE OF TECHNOLOGY PUNE

Presentation route

OUTLINE

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The deployment problem starts after rideshare reaches orbit

INTRODUCTION

Shared destination

Secondary CubeSats inherit the orbit chosen for the primary customer.

Conventional release

Spring deployers separate payloads at about 1–2 m/s; release timing helps in-track phasing.

VOLLEY question

Can commanded release provide useful orbital choice at an acceptable installed cost?

Mission split: host = coarse orbital repositioning • deployer = each satellite's release condition

Existing methods solve different parts of the problem

LITERATURE REVIEW

Approach	Strength	Limit for this thesis
Spring deployer	Flight heritage; compact; simple	Limited tailored release speed
Timed release / drag	In-track phase without a high impulse	Does not quickly change orbital energy
Orbital transfer vehicle	Larger orbit changes	Greater bus, mass and integration burden
Electromagnetic research	Commandable acceleration studied	Armature, high-g or maturity constraints vary

A narrower, testable contribution

RESEARCH GAP

Commanded release for an unmodified CubeSat at moderate acceleration, with the full system counted.

Contribution examined

Gen5 reusable-sled linear motor, 3U payload, magazine, pulse power, arrest and orbital utility.

Boundary

Electromagnetic CubeSat deployment exists in prior work. Payload qualification and physical performance remain unproved.

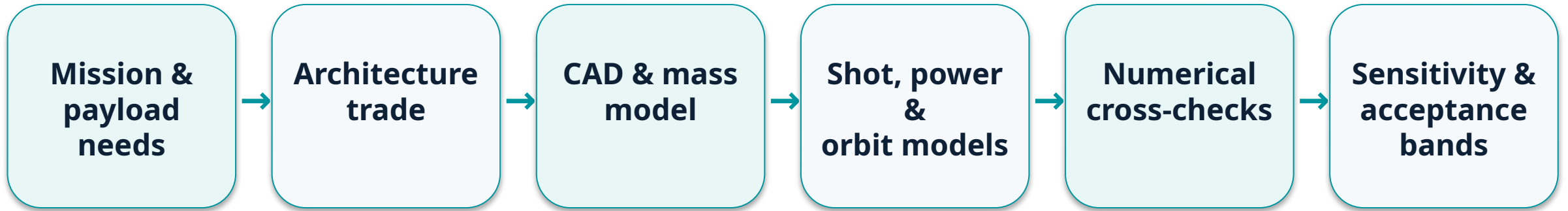
What was achieved, and what remains open

OBJECTIVES

Objective	Evidence at final review
Architecture for an unmodified 3U	Gen5 CAD/model; payload qualification open
Quantify shot and commanded speed	Shot/circuit/control simulations; no measured shot
Quantify orbital utility	Modelled benefit and numerical cross-checks; atmospheric limits
Compare full system burden	Mass and alternatives analysed; 3U mass criterion fails
Define next test path	Gen6 gates set; mechanism and build release open

Requirements → model → challenge → revise

RESEARCH METHODOLOGY



**Numerical cross-checks verify particular models.
Experimental validation has not yet begun.**

The models answer specific questions, with specific limits

RESEARCH METHODOLOGY

Question	Method / cross-check	Evidence limit
Field and thrust	Analytic winding; magpylib; FEM	No measured thrust constant
Shot and power	Dynamics; Monte Carlo; ngspice	Assumed components and interfaces
Structure and flow	CAD; CalculiX; OpenFOAM	No payload qualification
Orbital change	Orbit model; Cowell / GMAT cases	Lifetime depends on atmosphere

Project evolution: documented decisions and analysis

TIMELINE

2023

Hosted concept

Free-flyer concept reframed around a host platform

Mid-2025*

LSM direction

Coilgun study gives way to a linear synchronous motor

2025–26*

CAD generations

Geometry and mass models mature through Gen5

Jul–Sep 2026

Gen5 evidence

Cross-checks, defect corrections and thesis record

Sep 2026 →

Gen6 trade

Mechanism choice and mission envelope reopened

* Approximate historical milestone; not a documented semester deadline.

Next work is gated by evidence, not an invented date

TIMELINE

N0

Freeze mission, payload and host envelope

N1

Run coupled multi-payload mission cases

N2

Reject or select mechanisms on common criteria

N3–N4

Close installed burden, faults and BOLLEY drive

N5

Prepare first-test package and review

A mature analytical record; physical evidence still needed

WORK PROGRESS

Produced

Manuscript, CAD, scripts and figures, validation sheets, provenance and defect register.

Analysed

Gen5 shot, circuit, mass, thermal, structural and orbit cases. Failed criteria are retained.

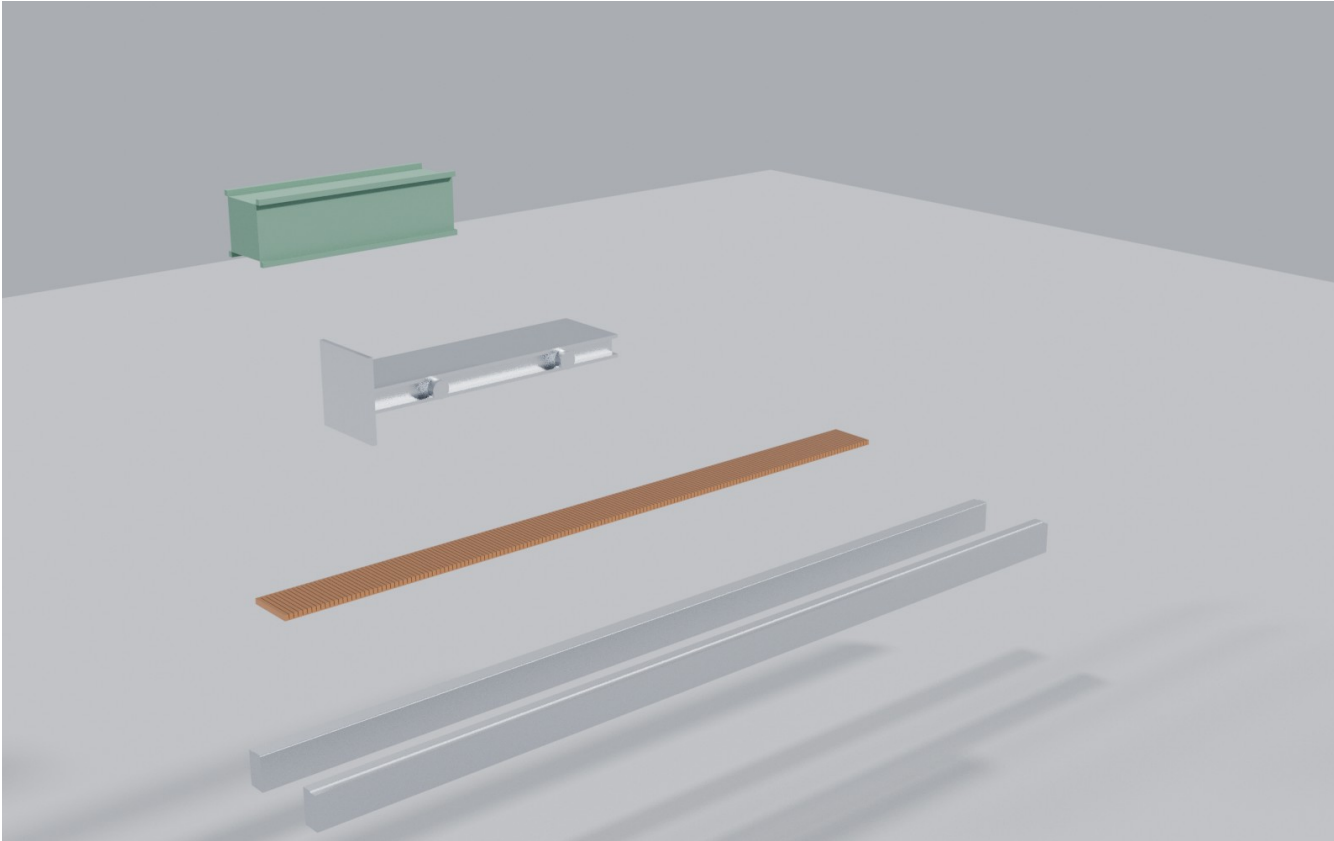
Still open

Hardware tests, provider interfaces, payload qualification, measured reliability and build release.

The presentation does not assign a percentage complete from document count.

Gen5: complete computational system configuration

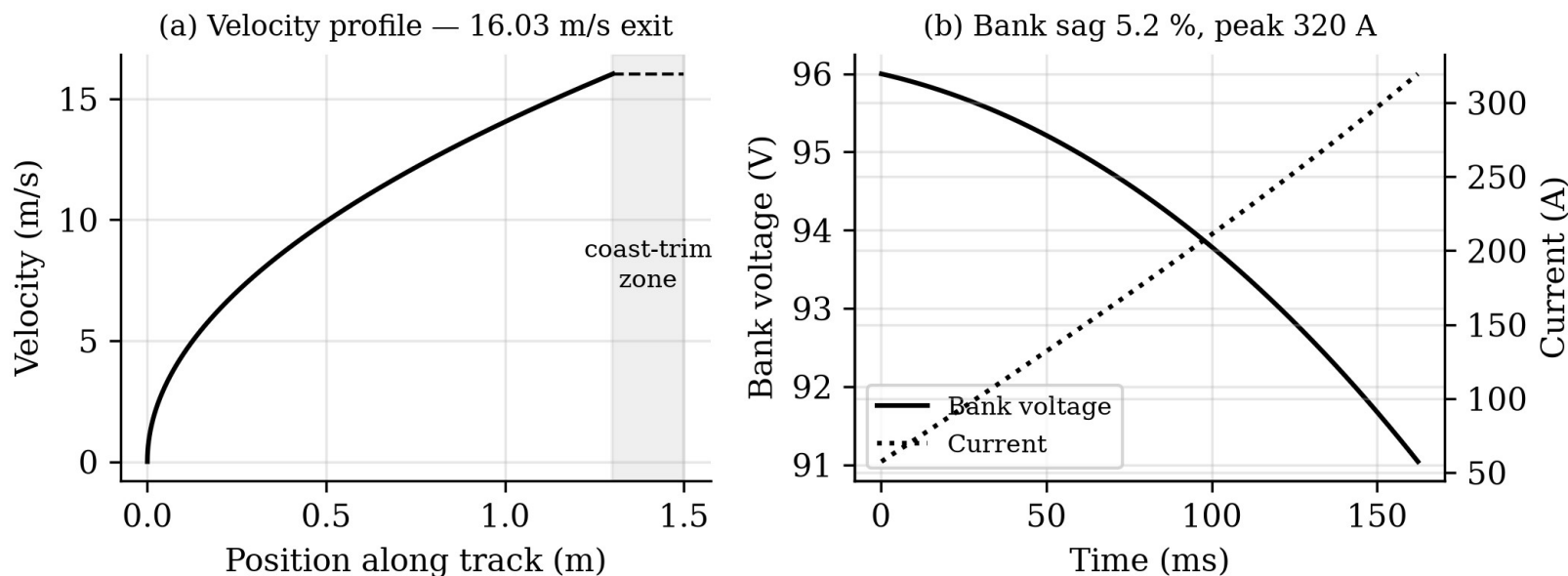
RESULTS



- 1.5 m linear-motor track and double-sided Halbach arrays
- Reusable 9.45 kg magnet sled
- Two cassettes; twelve 3U satellites
- Pulse bank, restraint and eddy-current arrest

Rated 3U shot: performance predicted by the model

RESULTS



16.0 m/s
exit speed

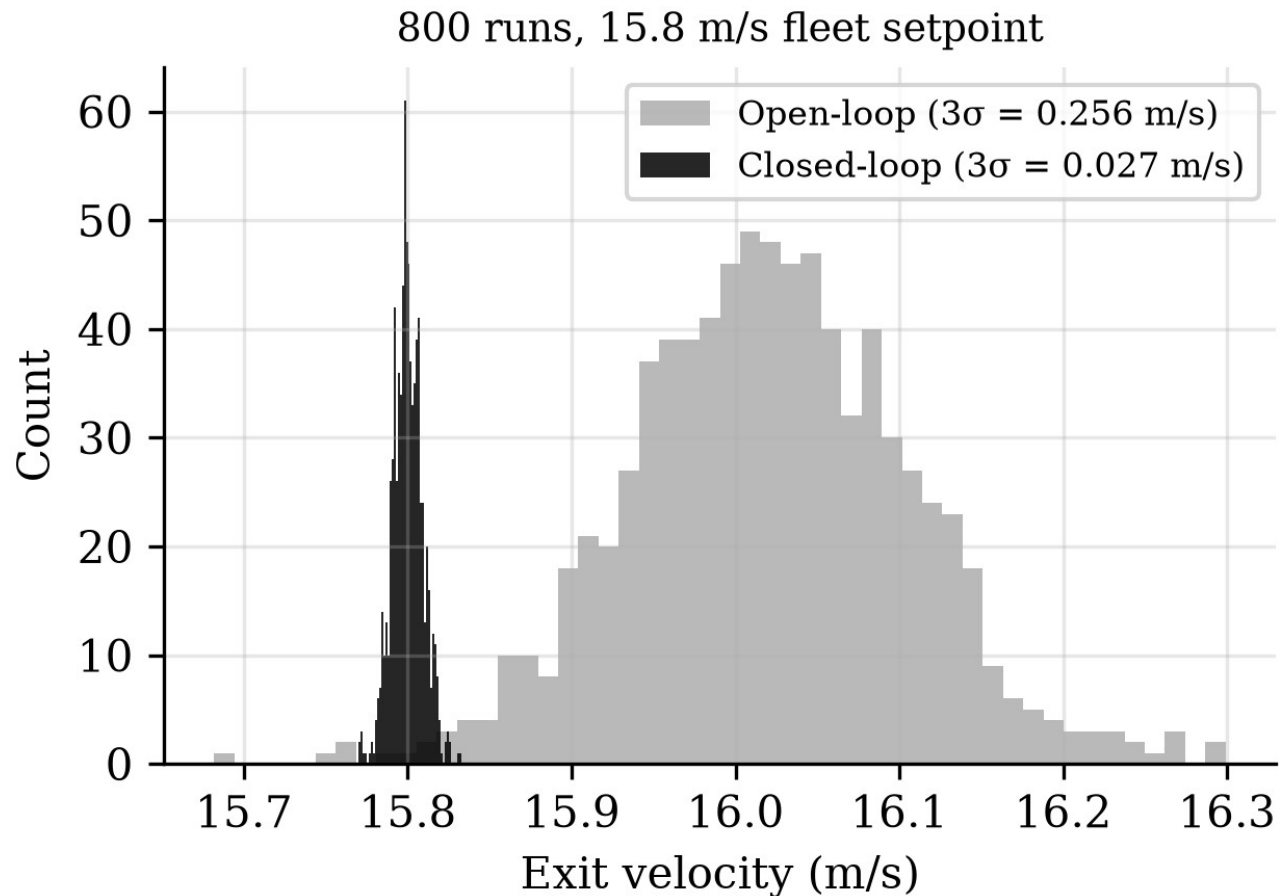
10.07 g
acceleration

2.782 kJ
gross draw

18.8%
net efficiency

Modelled command precision is narrow; it is unmeasured

RESULTS



0.0274 m/s

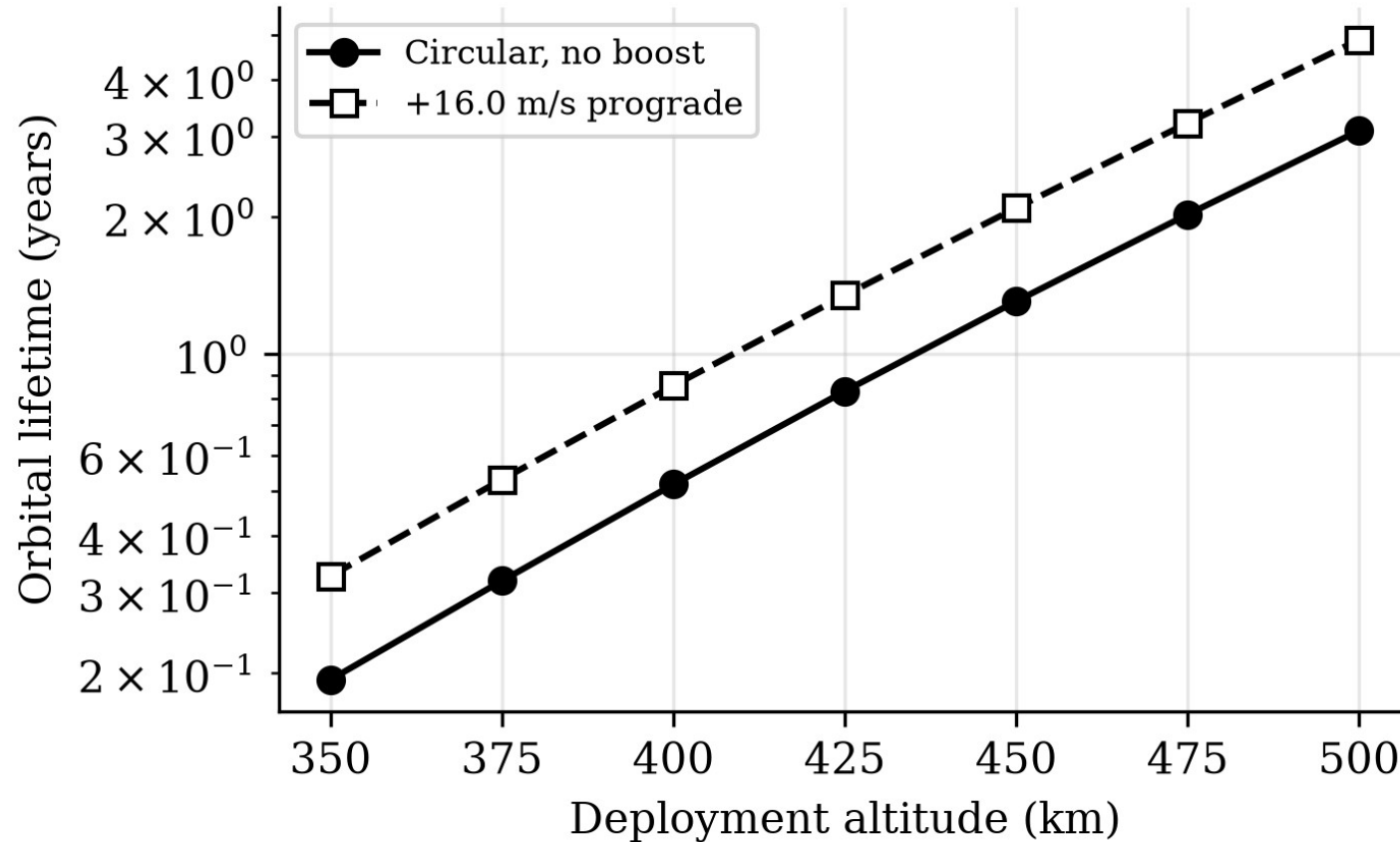
3σ exit-velocity dispersion at a 15.8 m/s fleet setpoint.

Evidence limit

Monte Carlo under modelled sensing and plant tolerances; no measured dispersion.

Commanded velocity can change orbital energy

RESULTS



+28.8 km

Predicted semi-major-axis rise for one maximum-velocity release.

1.60×

Predicted lifetime at stated mean solar activity; atmosphere-dependent.

Timing alone can create in-track phase; this result concerns orbital energy.

Gen5 fails its 3U mass comparison

RESULTS



1.76× heavier per 3U payload; the preset 15% parity band fails.

The unresolved physical questions determine viability

RESULTS

Payload interface

Arrest loads, contact and magnetic cleanliness need payload-specific evaluation.

Shared reliability

A failed retention or reload element can affect the remaining manifest.

Host compatibility

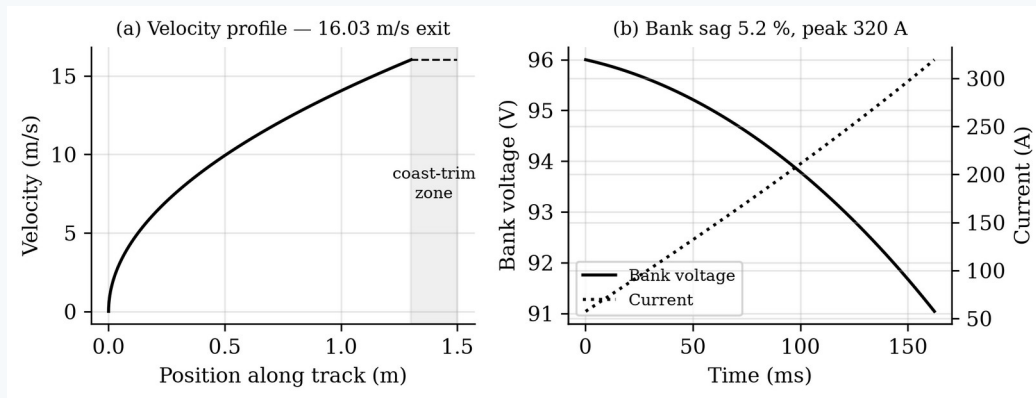
No launch provider has supplied an approved accommodation or operating interface.

FEM, CFD and CAD are numerical evidence. No prototype has been built or tested.

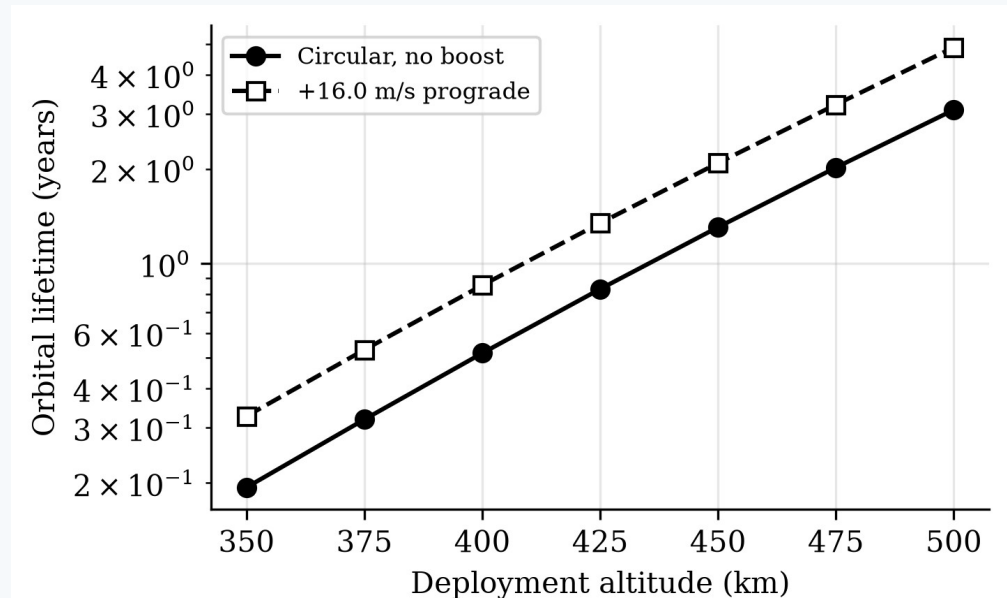
Computational demonstration: from shot to orbit

RESULTS

1. Simulate a 3U shot → 16.0 m/s



2. Model the orbital response



Compare predicted orbital change and full installed mass with the spring baseline. Simulation, not physical demonstration.

What this final review can defend

CONCLUSION

- Gen5 predicts commandable 16.0 m/s deployment for a 3U payload and a case-dependent orbital benefit.
- The modelled full system is 76% heavier per 3U customer than the spring comparator.
- Numerical checks strengthen specific models, while hardware behavior and interfaces remain unmeasured.
- Gen5 is a documented design study; the next configuration must earn a test-article decision.

Gen6 reopens the mechanism and installed-system trade

FUTURE WORK

One reusable path • sequential supported CubeSat sizes • individually commanded release

- Investigate approximately 1–2 m/s through 1 km/s across mission cases.
- Compare electromagnetic, stored-energy, gas/fluid, hybrid, BOLLEY and conventional options.
- Count feeder, retention, arrest, control, energy, thermal, host loads and shared faults.

1 km/s is a research target, not demonstrated speed or payload/provider compatibility.

Next milestone: one reviewable first-test configuration

FUTURE WORK

Freeze

Named mission, payload and host surrogate

Decide

Fair mechanism and installed-burden trade

Measure

Field, contact, release, arrest, loads and cycle behavior

Review

Drawings, BOM, procedures, uncertainty and review decision

Core literature and standards (1/2)

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- [4] H. Feng et al., 'Design and reachable domain analysis of on-orbit electromagnetic launcher for CubeSats,' Int. J. Aerospace Eng., 2025.
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Methods and project records (2/2)

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